

MAT 549 HW06

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Due 04/21

Problem 1 (16-3(a)). Suppose E and M are smooth n -manifolds with or without boundary, and $\pi: E \rightarrow M$ is a smooth k -sheeted covering map or generalized covering map. Show that if E and M are oriented and π is orientation-preserving, then

$$\int_E \pi^* \omega = k \int_M \omega$$

for any compactly supported n -form ω on M .

Solution. Let $\mathcal{U}$ be an open cover of M by connected evenly covered open sets and let $\{\varphi_U : U \in \mathcal{U}\}$ be a partition of unity subordinate to this open cover. Then, $\tilde{\mathcal{U}} = \{\tilde{U} = \pi^{-1}(U) : U \in \mathcal{U}\}$ is an open cover of E and the functions $\psi_{\tilde{U}} = \varphi_U \circ \pi$ for $\tilde{U} \in \tilde{\mathcal{U}}$ form a partition of unity on E subordinate to $\tilde{\mathcal{U}}$. To see that this collection of functions is indeed a partition of unity, we of course have that $\psi_{\tilde{U}}(x) = \varphi_U(\pi(x)) \in [0, 1]$ and

$$\sum_{\substack{\tilde{U} \in \tilde{\mathcal{U}} \\ x \in \tilde{U}}} \psi_{\tilde{U}}(x) = \sum_{\substack{U \in \mathcal{U} \\ \pi(x) \in U}} \varphi_U(\pi(x)) = 1$$

for all $x \in E$. Thus

$$\int_E \pi^* \omega = \sum_{\tilde{U} \in \tilde{\mathcal{U}}} \int_{\tilde{U}} \psi_{\tilde{U}} \cdot \pi^* \omega = \sum_{\tilde{U} \in \tilde{\mathcal{U}}} \int_U \sum_{i=1}^k \varphi_U \cdot \omega = k \sum_{U \in \mathcal{U}} \int_U \varphi_U \cdot \omega = k \int_M \omega,$$

where the second equality is due to the fact that $\pi^{-1}(U)$ consists of k components diffeomorphic to U via the orientation-preserving map π and that $\psi_{\tilde{U}} = \varphi_U \circ \pi$. $\square$

Problem 2 (16-4). Suppose M is an oriented compact smooth manifold with boundary. Show that there does not exist a smooth retraction of M onto its boundary.

Solution. Suppose a smooth retraction $r: M \rightarrow \partial M$ exists. Since M is orientable, so is its boundary, so let ω be an orientation form on ∂M , which in particular means that $\int_{\partial M} \omega \neq 0$. Then, Stokes's theorem gives the following contradiction:

$$\int_{\partial M} \omega = \int_{\partial M} r^* \omega = \int_M d(r^* \omega) = \int_M r^*(d\omega) = 0.$$

To be thorough, the first equality holds because the pullback of $r^* \omega$ by the inclusion $\partial M \hookrightarrow M$ gives ω because r composed with the inclusion is the identity and the last equality is true because ω is a top-degree form on ∂M , hence $d\omega = 0$. $\square$

Problem 3 (16-5). Suppose M and N are oriented, compact, and connected smooth manifolds, and $F, G: M \rightarrow N$ are homotopic diffeomorphisms. Show that F and G are either both orientation-preserving or both orientation-reversing.

Solution. Let ω be an orientation form on N , $I = [0, 1]$, and let $H: M \times I \rightarrow N$ be a homotopy from F to G , which we can take to be smooth by Theorem 6.29 and which satisfies $H(p, 0) = F(p)$ and $H(p, 1) = G(p)$ for all $p \in M$. By Stokes's theorem and the fact that $d\omega = 0$,

$$\int_{\partial(M \times I)} H^* \omega = \int_{M \times I} dH^* \omega = \int_{M \times I} H^* d\omega = 0.$$

Then, $\partial(M \times I) = M \times \{0, 1\}$ with the map $i: M \times \{0\} \rightarrow M \times \{1\}$ given by $i(x, 0) = (x, 1)$ being orientation-reversing (due to the orientation that $\partial I = \{0, 1\}$ inherits from I). Then, if $\iota_k: M \rightarrow M \times \{k\}$ is the map $\iota_k(x) = (x, k)$ for $k \in \{0, 1\}$,

$$\begin{aligned} 0 &= \int_{M \times I} H^* \omega = \int_{M \times \{0\}} H^* \omega - \int_{M \times \{1\}} H^* d\omega = \int_M \iota_0^* H^* \omega - \iota_1^* H^* \omega \\ &= \int_M (H \circ \iota_0)^* \omega - (H \circ \iota_1)^* \omega = \int_M F^* \omega - G^* \omega, \end{aligned}$$

which implies that F and G either both preserve or both reverse orientation (since $\int_M F^* \omega = \int_M G^* \omega = \pm \int_M \omega$, with the sign depending on whether or not F and G reverse orientation). $\square$

Problem 4 (16-6). Prove that the following are equivalent.

- (a) There exists a nowhere-vanishing vector field on S^n .
- (b) There exists a continuous map $V: S^n \rightarrow S^n$ satisfying $V(x) \perp x$ for all $x \in S^n$.
- (c) The antipodal map $\alpha: S^n \rightarrow S^n$ is homotopic to the identity map.
- (d) The antipodal map $\alpha: S^n \rightarrow S^n$ is orientation-preserving.
- (e) n is odd.

Solution. (a) implies (b). This follows from the basic fact of Euclidean geometry that the unit normal at a given point $x \in S^n$ is x : if $\hat{V}$ is a nowhere zero vector field on S^n , then let $V: S^n \rightarrow S^n$ be defined by $V(x) = \frac{\hat{V}(x)}{\|\hat{V}(x)\|}$, so that $V(x)$ is tangent to S^n at x , hence orthogonal to x .

(b) implies (c). Let $H: S^n \times I \rightarrow S^n$ be the map

$$H(x, t) = \cos(\pi t)x + \sin(\pi t)V(x),$$

which is well-defined because $\|H(x, t)\| = \cos(\pi t)^2 + \sin(\pi t)^2 = 1$ for all $(x, t) \in S^n \times I$ and satisfies $H|_{S^n \times \{0\}} = \text{Id}_{S^n}$, $H|_{S^n \times \{1\}} = \alpha$, so H is a homotopy between α and the identity map.

(c) implies (d). This follows from Problem 16-5 and the fact that the identity map is obviously orientation-preserving.

(d) implies (e). This follows from Problem 15-3 from last week's homework.

(e) implies (a). For $n = 2k - 1$ odd, let V be the vector field on $S^n \subseteq \mathbb{R}^{2k}$ defined by

$$V(x_1, y_1, x_2, y_2, \dots, x_k, y_k) = (-y_1, x_1, -y_2, x_2, \dots, -y_k, x_k),$$

which is a tangent vector field because it satisfies $V(x) \perp x$ for all $x \in S^n$, and is nowhere zero because $\|V(x)\| = \|x\| = 1$ for all $x \in S^n$. $\square$